

Experiment-1

1 Develop a program to blink 5 LEDs back and forth.

Step-by-Step Instructions:

STEP 1: Log in to Simulator /Tinker cad tool.

STEP 2: Select the Circuits option.

STEP 3: Select Create new circuit option.

STEP 4: Drag/ Select Arduino Uno R3 to the workspace.

STEP 5: Drag Breadboard small to the workspace.

STEP 6: Drag five LEDs one by one, and change the color.

STEP 7: Connect the LEDs to Arduino using a breadboard.

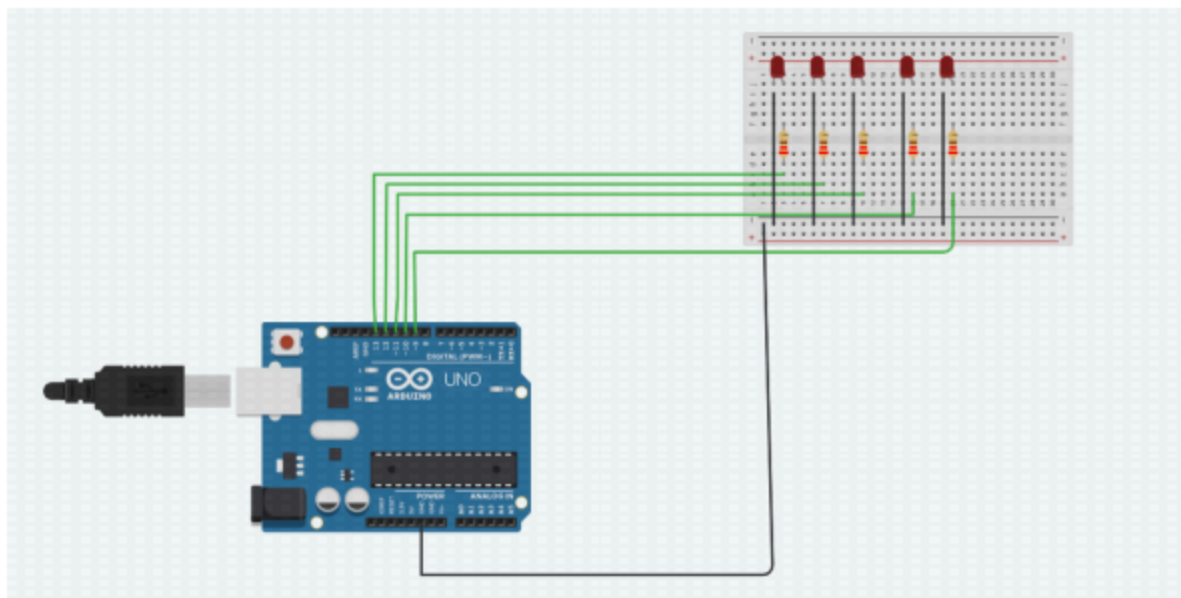
STEP 8: Connect the GND pin of Arduino to the negative line of breadboard through the 220 Ω Resistor. STEP 9: Connect all cathode lines of LEDs to the negative line of breadboard vertically.

STEP 10: Connect the Anode/ tilted pin of LEDs to Arduino pins

Components:

- Arduino Uno
- Breadboard
- 5 LEDs
- 5 resistors (220 Ω)
- Jumper wires

Circuit Diagram



Source Code :

```
#define led1 13
#define led2 12
#define led3 11
#define led4 10
#define led5 9

int leds[] = {13,12,11,10,9}; // LEDs connected to pins 13-9 pins
int numLeds = 5;
int delayTime = 200; // milliseconds

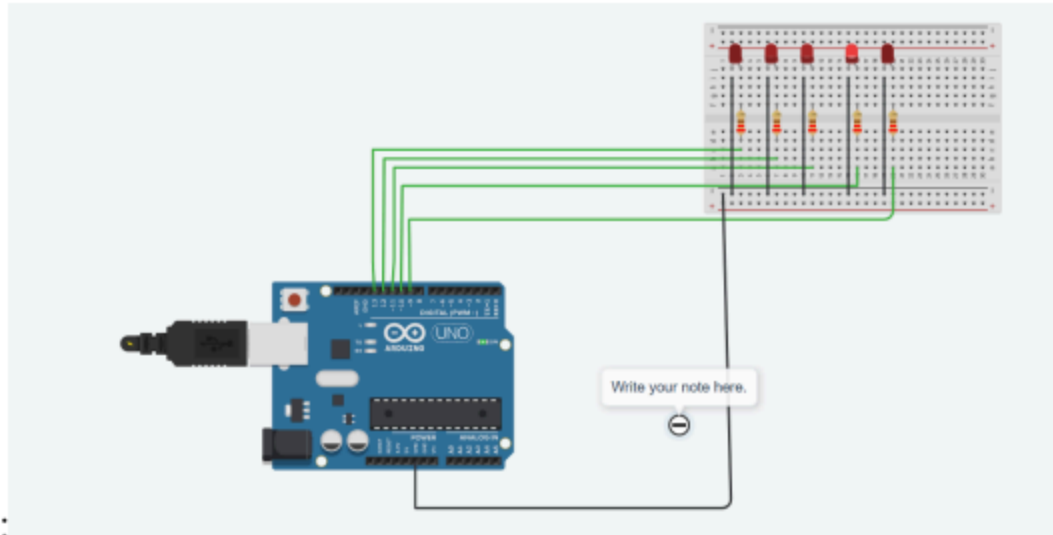
void setup() {
  for (int i = 0; i < numLeds; i++) {
    pinMode(leds[i], OUTPUT);
  }
}

void loop() {
  // Move left to right
  for (int i = 0; i < numLeds; i++) {
    digitalWrite(leds[i], HIGH);
    delay(delayTime);
    digitalWrite(leds[i], LOW);
  }

  // Move right to left
  for (int i = numLeds - 2; i > 0; i--) {
    digitalWrite(leds[i], HIGH);
    delay(delayTime);
    digitalWrite(leds[i], LOW);
  }
}
```



OUTPUT



1. Turn on LEDs left to right.
2. Turn off LEDs left to right.
3. Turn on LEDs right to left.
4. Turn off LEDs right to left.



Experiment-2

Develop a program to interface a Relay with Aurdino Board

STEP 1: Log in to Simulator.

STEP 2: Select the Circuits option.

STEP 3: Select Create new circuit option.

STEP 4: Drag/ Select Arduino Uno R3 to the workspace.

STEP 5: Drag Relay to the workspace.

STEP 6: Drag Bulb to the workspace.

STEP 7: Drag Power supply to the workspace.

STEP 8: Connect negative of power supply to terminal 1 of bulb.

STEP 9: Connect positive of power supply to terminal 7 of relay.

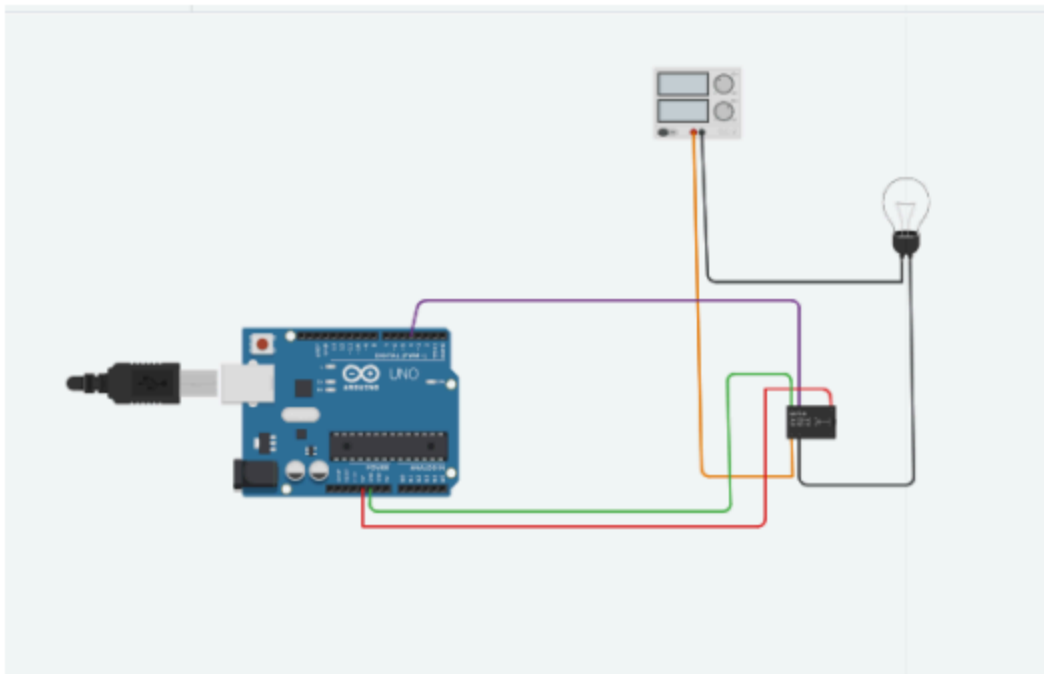
STEP 10: Connect terminal 8 of relay to terminal 2 of bulb.

STEP 11: Connect pin 3 of Arduino to terminal 5 of relay.

STEP 12: Connect 5V pin of Arduino to terminal 1 of relay.

STEP 13: Connect GND pin of Arduino to terminal 6 of relay.

Circuit Diagram



Code:

```
int relayPin = 4;
void setup()
{
  pinMode(relayPin,OUTPUT);
}
void loop()
{
  digitalWrite(relayPin,HIGH);
  delay(1000); // Wait for 1000 millisecond(s)
  digitalWrite(relayPin,LOW);
  delay(1000); // Wait for 1000 millisecond(s)
}
```

Output :

Turn on and off the Bulb based on given delay.



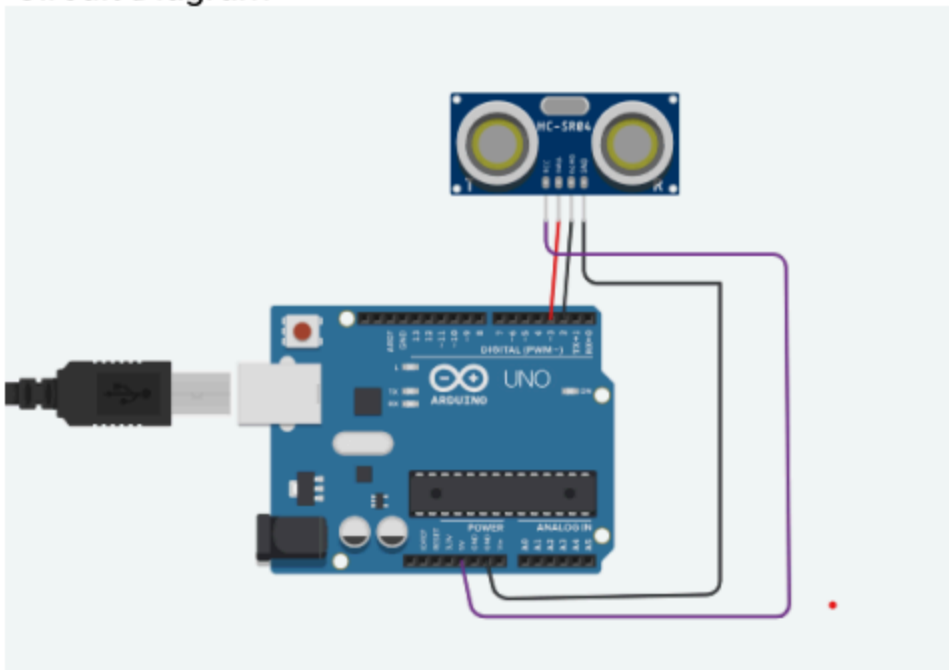
Experiment- 3

Develop a program to deploy an intrusion detection system using Ultrasonic and sound sensors.

Components Required

1. HC-SR04 ultrasonic sensor
 2. Aurdino Board
 3. Jumper wires
- Connect the circuit as per circuit.
 - Make sure VCC and Ground pins connected properly to avoid any damage to Arduino board.
 - Open Arduino IDE then goto tools and select appropriate Arduino board.

Circuit Diagram



CODE

```
#define trigPin 3  
#define echoPin 2
```

```
long duration;  
float distance;
```

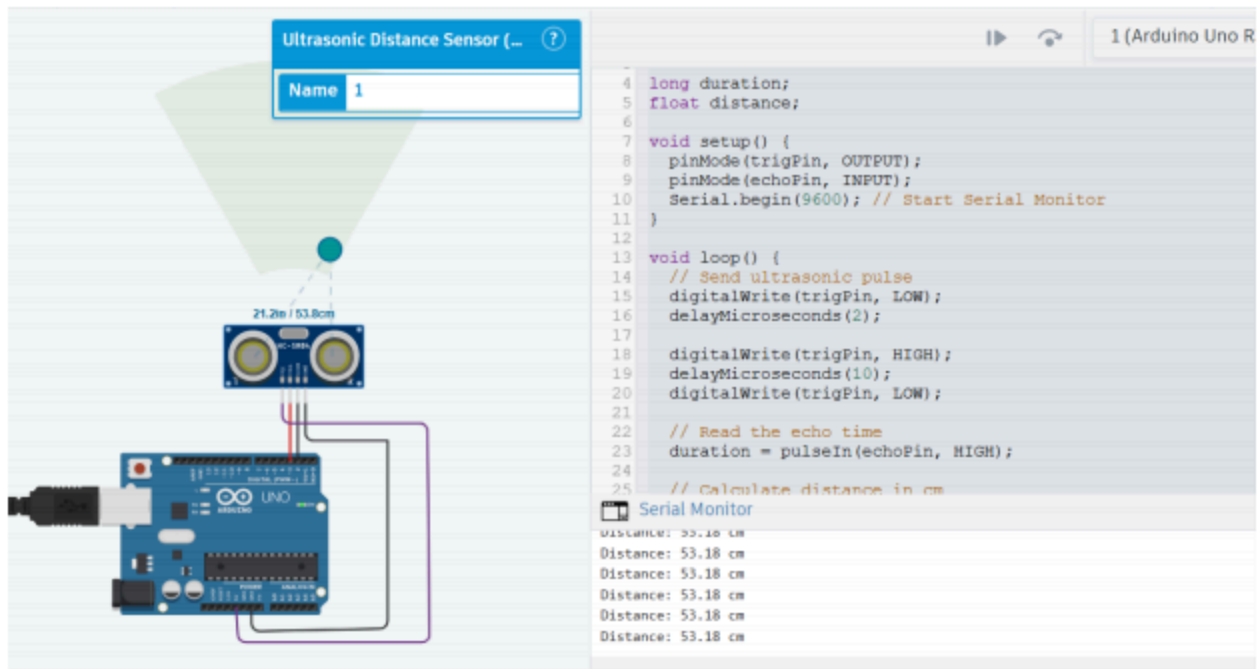


```
void setup() {  
  pinMode(trigPin, OUTPUT);  
  pinMode(echoPin, INPUT);  
  Serial.begin(9600); // Start Serial Monitor  
}
```

```
void loop() {  
  // Send ultrasonic pulse  
  digitalWrite(trigPin, LOW);  
  delayMicroseconds(2);  
  
  digitalWrite(trigPin, HIGH);  
  delayMicroseconds(10);  
  digitalWrite(trigPin, LOW);  
  
  // Read the echo time  
  duration = pulseIn(echoPin, HIGH);  
  
  // Calculate distance in cm  
  distance = duration * 0.034 / 2;  
  
  // Print to Serial Monitor  
  Serial.print("Distance: ");  
  Serial.print(distance);  
  Serial.println(" cm");  
  
  delay(500);  
}
```

OUTPUT:





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Experiment- 4

Develop a program to control a DC motor with Arduino board.

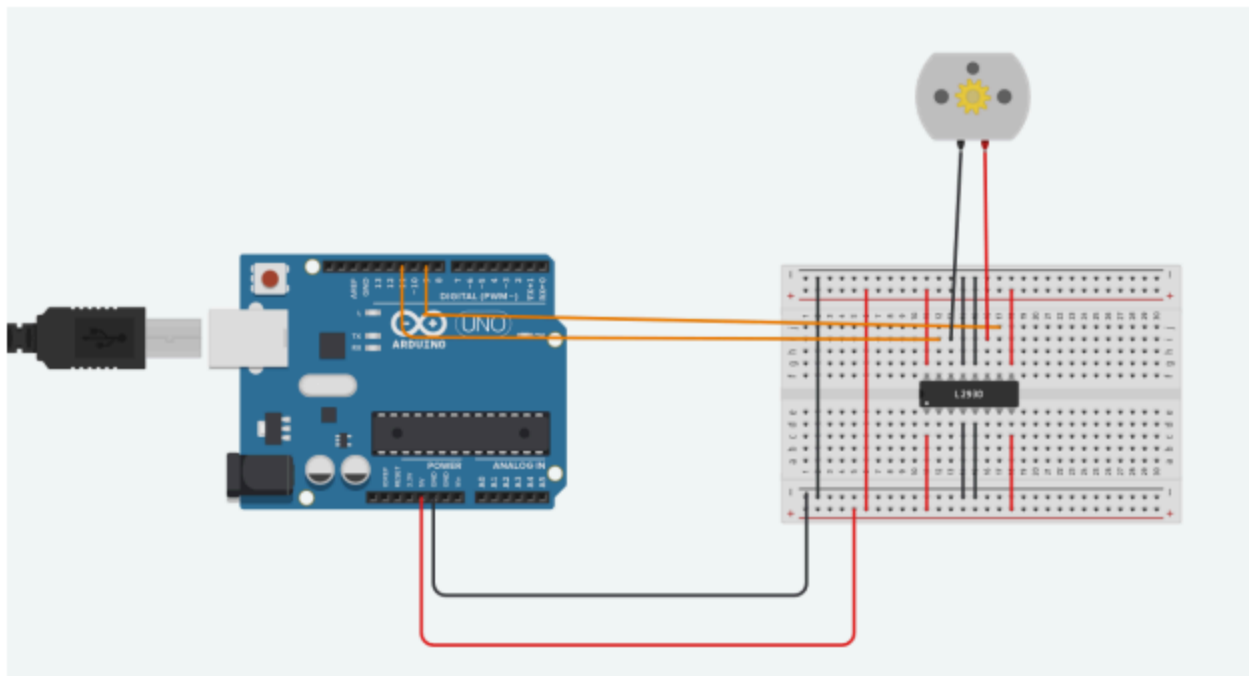
Components Required

- Arduino Uno
- L293D Motor Driver IC
- DC Motor
- Breadboard
- Jumper wires

Steps

- a) Connect the circuit as per circuit.
- b) Make sure VCC and Ground pins connected properly to avoid any damage to Arduino board.
- c) Open Arduino IDE then goto tools and select appropriate Arduino board.
- d) Select necessary components and make circuit connections as shown in the diagram
- e) write the code and run a simulator to get appropriate results.

Circuit Diagram:





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CODE:

```
void setup()
{
  pinMode(11, OUTPUT);
  pinMode(9, OUTPUT);
}

void loop()
{
  digitalWrite(11, HIGH);
  digitalWrite(9, LOW);
  delay(1000); // Wait for 1000 millisecond(s)

  digitalWrite(9, HIGH);
  digitalWrite(11, LOW);
  delay(1000); // Wait for 1000 millisecond(s)
}
```

OUTPUT

Successfully demonstrated control a DC motor with Arduino board



Experiment- 5

Develop a program to deploy smart street light system using LDR

Components required:

1. Arduino Uno – 1
2. Breadboard – 1
3. Jumper Wires – 4
4. LED – 1
5. Resistors ($850\ \Omega$ / $1k\Omega$) – 2
6. LDR – 1

An LDR is a light-sensitive device **whose** resistance changes with the intensity of light **falling on it**.

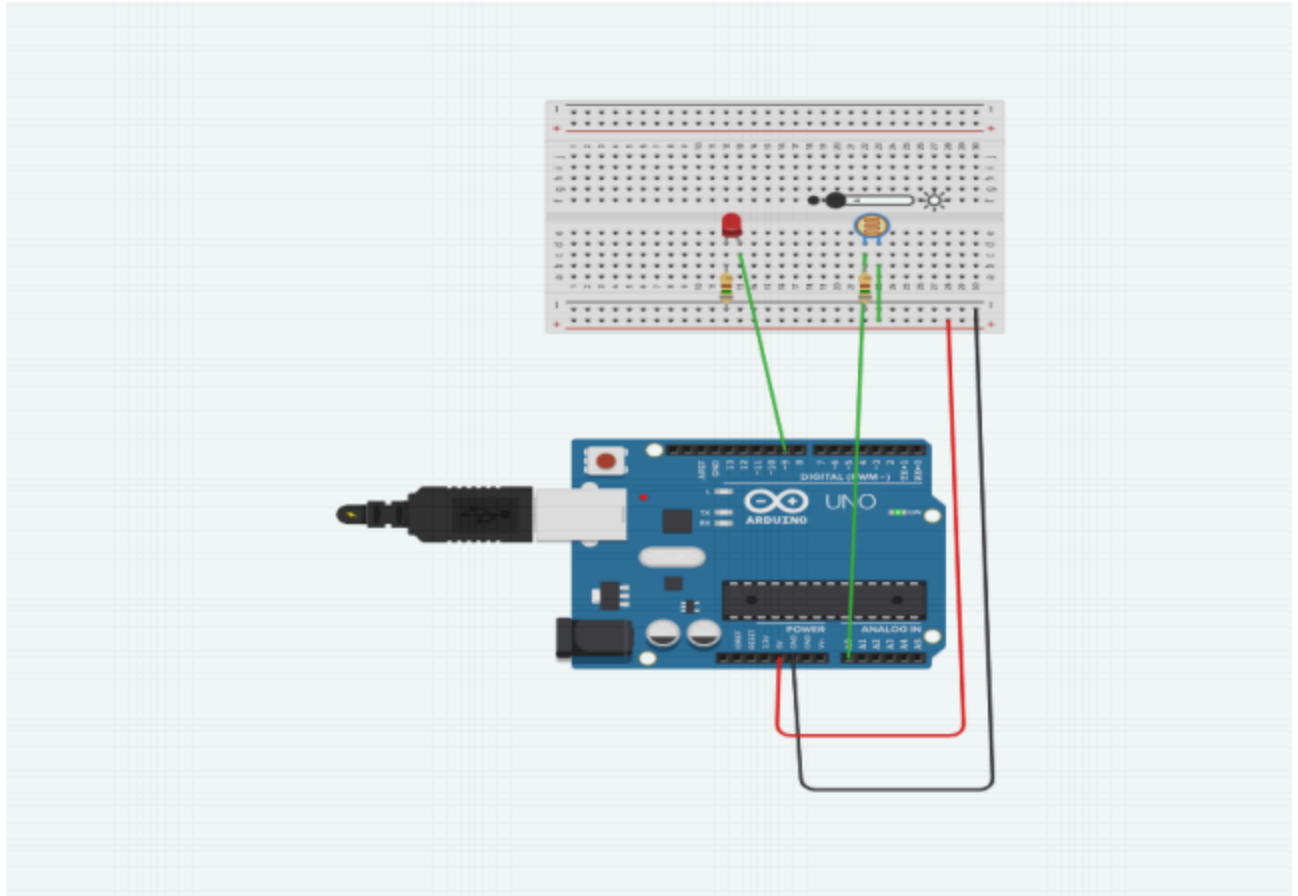
- In bright light → LDR resistance is LOW → LED is OFF
- In darkness → LDR resistance is HIGH → LED is ON

STEPS

- a) Connect the circuit as per the circuit diagram.
- b) Select appropriate components .
- c) Ensure **VCC and GND** pins are connected properly to avoid any damage to Arduino.
- d) Open Arduino IDE → go to **Tools** and select the correct Arduino board.
- e) Type the program (sketch) and upload it to the board.
- f) Run the simulation

Circuit conection





Code

```
int sensorPin = A0;  
int sensorValue = 0;  
int led = 9;  
  
void setup() {  
  pinMode(led, OUTPUT);
```



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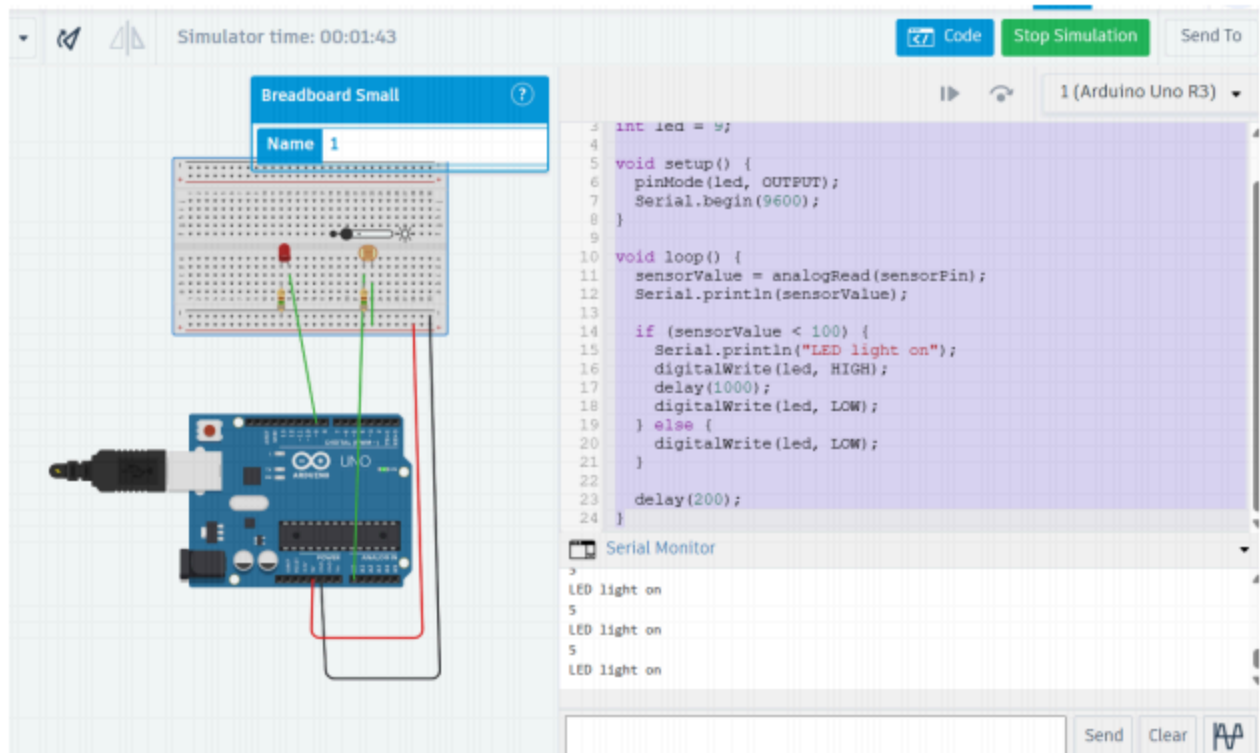
```
Serial.begin(9600);  
  
}  
  
void loop() {  
  sensorValue = analogRead(sensorPin);  
  Serial.println(sensorValue);  
  
  if (sensorValue < 100) {  
    Serial.println("LED light on");  
    digitalWrite(led, HIGH);  
    delay(1000);  
    digitalWrite(led, LOW);  
  } else {  
    digitalWrite(led, LOW);  
  }  
  
  delay(200);  
}
```

OUTPUT:

Click on LDR sensor ,you will notice light intensity adjusting components.

Adjust the light intensity and check the values in Serial monitors





Experiment- 6

Develop a program to classify dry and wet waste with the Moisture sensor (DHT22).

Components Needed in Tinkercad:

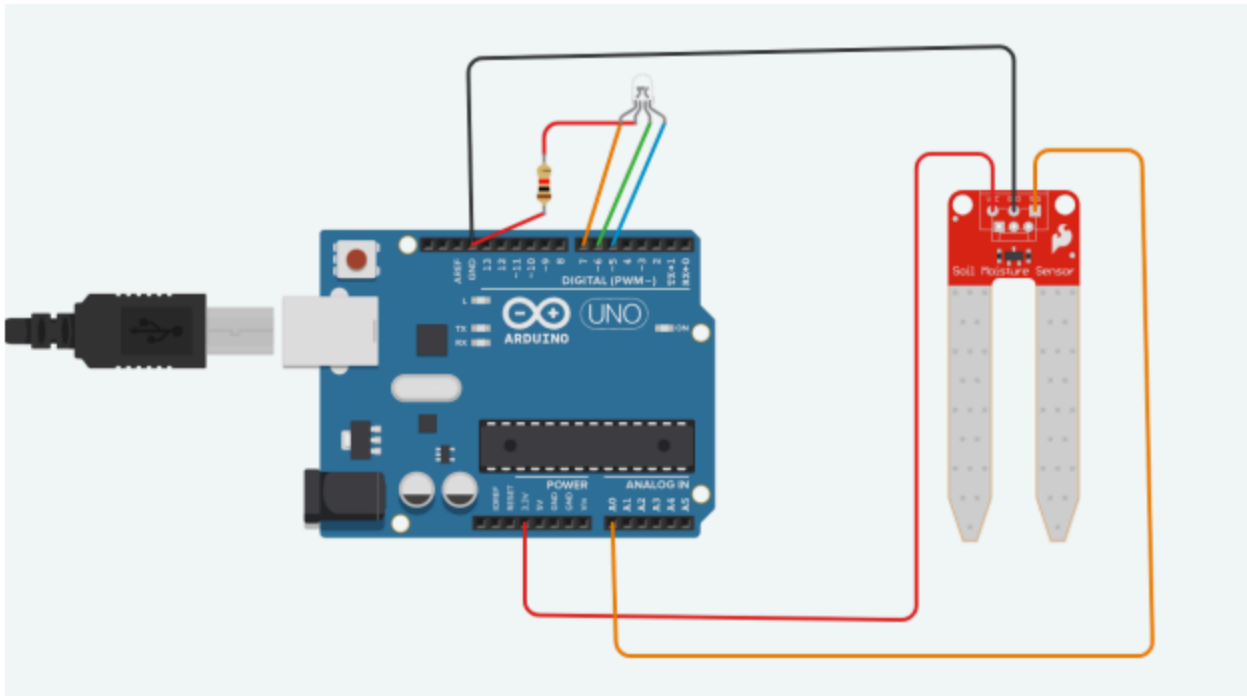
1. Arduino UNO
2. DHT22 sensor (use as humidity sensor)
3. RGB Led (optional) Serial Monitor for output
4. Breadboard and wires
5. Resistor 1k ohm

STEPS

- a) Connect the circuit as per circuit.
- b) Make sure VCC and Ground pins connected properly to avoid any damage to Arduino board.
- c) Open Arduino IDE then goto tools and select appropriate Arduino board.
- d) Select necessary components and make circuit connections as shown in the diagram
- e) write the code and run a simulator to get appropriate results.



Circuit Diagram



CODE

```
int moistureValue;  
  
float moisture_percentage;  
  
void setup()  
{  
  pinMode(7, OUTPUT);  
  pinMode(6, OUTPUT);  
  pinMode(5, OUTPUT);  
  Serial.begin(9600);  
}  
  
void loop()  
{  
  moistureValue = analogRead(A0);
```

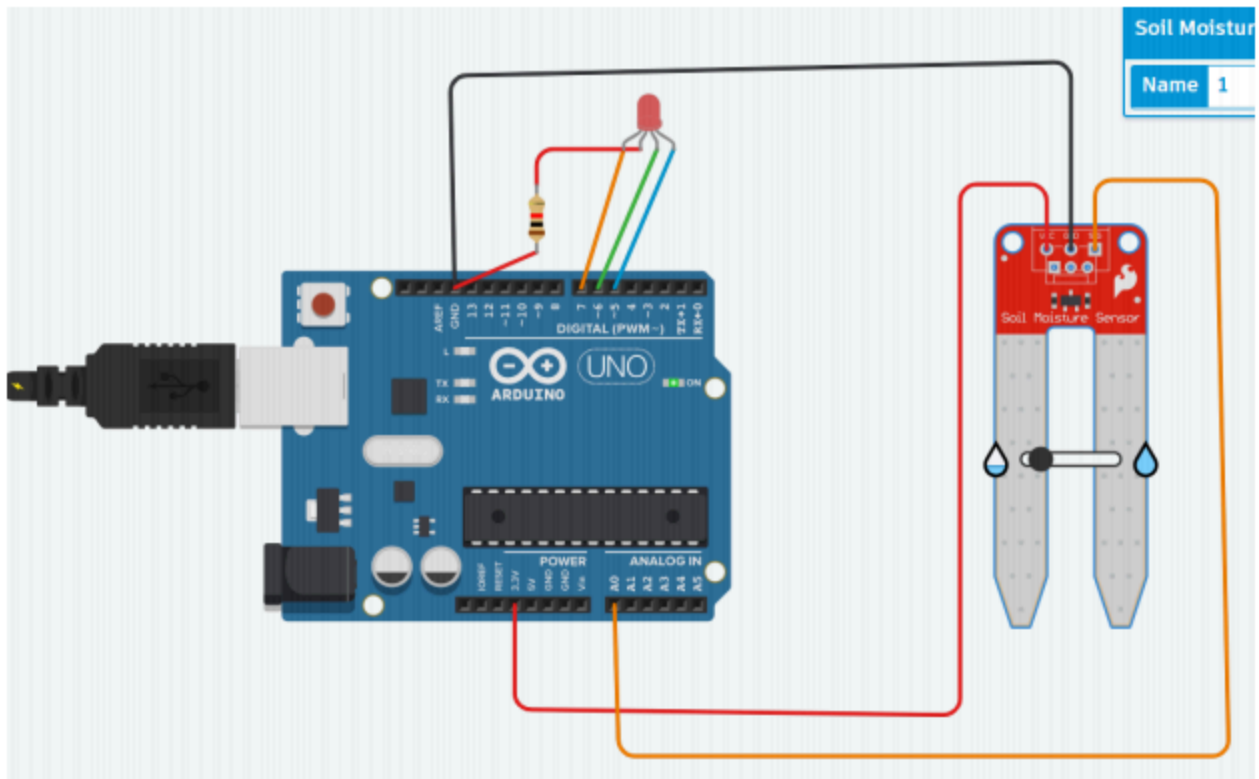


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```
moisture_percentage = ((moistureValue/539.00)*100);  
if ( moisture_percentage>0 && moisture_percentage<25 )  
{ digitalWrite(7,HIGH);  
  digitalWrite(6,LOW);  
  digitalWrite(5,LOW);  
}  
if ( moisture_percentage>25 && moisture_percentage<80 )  
{ digitalWrite(7,LOW);  
  digitalWrite(6,HIGH);  
  digitalWrite(5,LOW);  
}  
if ( moisture_percentage>80 && moisture_percentage<100 )  
{ digitalWrite(7,LOW);  
  digitalWrite(6,LOW);  
  digitalWrite(5,HIGH);  
}  
Serial.print("\nMoisture Value : ");  
Serial.print(moisture_percentage);  
Serial.print("%");  
delay(1000);  
}
```

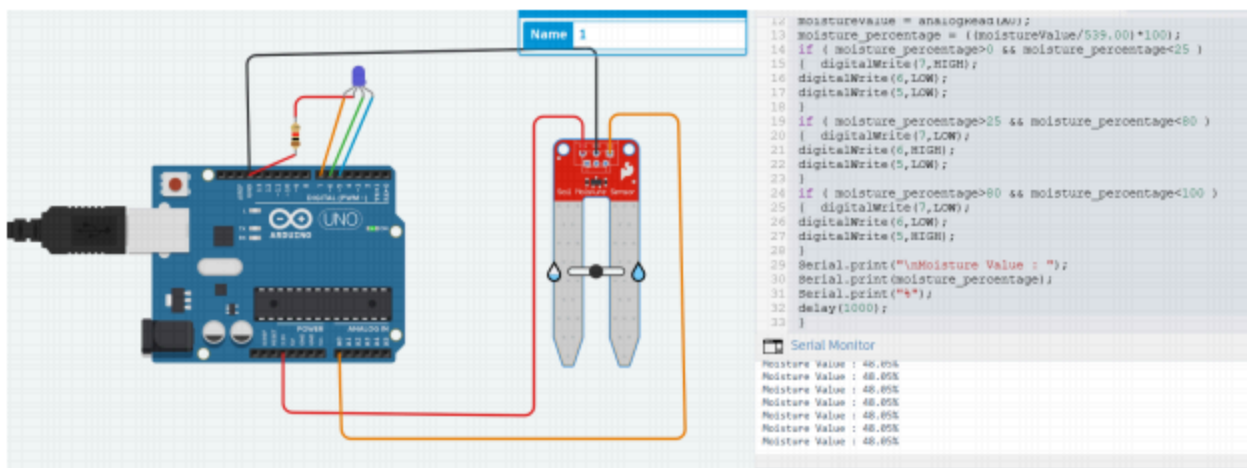
OUTPUT





If (moisture_percentage>0 && moisture_percentage<25)

Dry waste-Red led light switches on

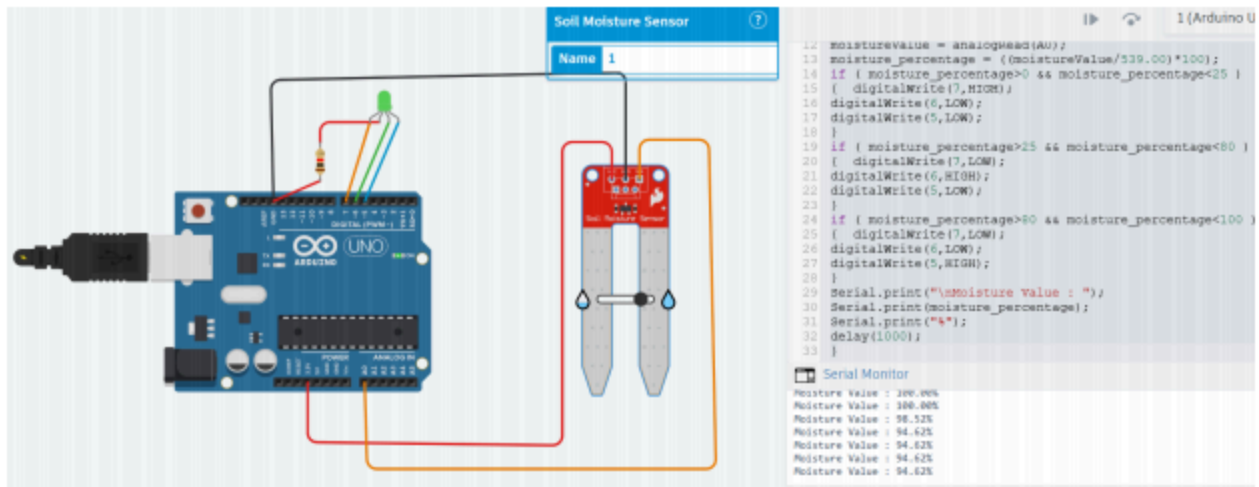


If (moisture_percentage>25 && moisture_percentage<80)

Blue light switches on indicating medium moisture in waste



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If (moisture_percentage>80&& moisture_percentage<100)

Green led light switches on indicating wet waste .



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